

Blockchains in Supply Chains: Potential Research Directions

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Abstract— The blockchain is a distributed digital ledger technology that is maintained and governed by a consensus mechanism. While the technology took root in the financial services industry, supply chain and logistics industry are considered to be the next promising use case for this fast-evolving technology. This paper recaps, reviews, and discusses key research issues that needs to be explored pertinent to application of blockchains in supply chains and logistics.

Keywords—*blockchain, supply chain, visibility, smart contracts*

I. INTRODUCTION

Blockchain technology is rapidly being integrated into many industries. One field of application that is of primary interest to us is supply chain, wherein blockchain technology could have a vast potential, particularly in areas such as supplier payments, product traceability, and contract bids and execution. Firms that adopt blockchain technology are expected to have more transparent supply chains and closer supplier-customer relationships that can result in better financial and operational performance. Blockchains can be leveraged to achieve these goals via integration with existing enterprise systems and augmenting technologies such as Internet of Things and RFID to monitor supply chain and logistics processes. There are a few examples of the adoption of blockchain technology within the supply chain and logistics domain. Recently, Accenture has announced a collaboration with Mastercard, Amazon Web Services, Everledger, and Mercy Corp to introduce a “circular supply chain” that allows consumers to make more sustainable buying choices [1]. Another example of blockchain adoption is Daimler, as part of their supply chain visibility initiative, announcing a partnership with Icertis, a leading provider of enterprise contract management solutions, to implement a blockchain solution for consistent documentation of contracts in the supply chain [2], [3].

As with any new technology, the gap between the theory and the practice is the main impediment to sophisticated decision-making regarding its adoption. It is therefore imperative that firms have a roadmap to gauge the suitability and feasibility of adopting the blockchain technology.

We focus on blockchain applications as an information-sharing tool in supply chains. If the incentives of its participants are aligned, i.e., if the risks and rewards of transactions are fairly distributed across the entire network, the supply chain operates

efficiently [4]. Firms can, contingent on better information sharing, align incentives across the supply chain by monitoring more business variables in order to avoid sub-optimal supply chain performance. Furthermore, information sharing can also be utilized to signal the quality of a product by providing the provenance of items being used in production that helps firms gain a strategic advantage over their competitors. This paper provides a roadmap that links such potential benefits of blockchain technology to a theoretical framework.

II. BLOCKCHAIN – AN OVERVIEW

A. What is blockchain?

“A blockchain is a new type of database that enables multiple parties to share the database and to be able to modify that in a safe and secure way even if they don’t trust each other.” Gideon Greenspan [5], [6].

Blockchain, also known as distributed ledger technology, is a cryptographically secure database that generates a digital log of transactions and is shared across either a public or a private network. As a database, a blockchain is secured by cryptographic keys, and the decentralized and distributed nature of this technology ensures there is no single point of failure concerning efficient flow of information among the users of this database. Public and private blockchains have similarities, namely, both are decentralized peer-to-peer networks governed by a consensus mechanism, and both provide certain guarantees on the immutability of the ledger, regardless of whether some participants are malicious. What differentiates public and private blockchains is “who is allowed to participate in the network, execute the *consensus* protocol and maintain the shared ledger [7].” In other words, their rules of governance and limits on privacy vary. Public blockchains are also known as *permissionless* blockchains, while private blockchains are also known as *permissioned* or *closed* blockchains. The most well-known public networks are the Bitcoin [8] and Ethereum [9] blockchains. A private blockchain, or a private network, would find utility in applications such as banking where privacy is of utmost concern to users. Hyperledger Fabric [10] is an example of a private blockchain.

The most critical attribute of a blockchain is, however, the democratization of the process of updating the database. Therefore, blockchain is a valuable option in a low-trust environment, which otherwise comprises multiple

intermediaries working off of their hidden incentives. However, it is important to note that a blockchain is only as good as the information input in it, i.e., it simply ensures the integrity of the individuals making a transaction by ensuring that they have the right combination of a public or private key.

B. Pros and Cons of Blockchains

Babich and Hilary [32] discuss the strengths (visibility, aggregation, validation, automation, and resiliency) and weaknesses (lack of privacy, lack of standardization, “garbage in, garbage out” (GIGO) problem, the black box effect, and inefficiency) of blockchains with respect to operations and supply chain management applications. Blockchains however do not completely eliminate the need for trust. In addition, an uncertain regulatory environment and perceived legal risks, along with privacy and confidentiality issues, are considered to be the main challenges for widespread blockchain adoption.

While the potential benefits associated to transparency and trust are clear, these benefits are also offset by the limitations discussed in the preceding paragraph. Therefore, the decision of adopting a blockchain requires rigorous evaluation as the benefits could be easily offset by the risks.

III. APPLICATIONS IN SUPPLY CHAINS

What is the value of blockchains in supply chains? While there has been considerable interest among industry professionals and academics in the potential of blockchain applications, including but not limited to enhanced information visibility and transparency, traceability, and security, the understanding regarding the tangible benefits is still limited.

The globally dispersed nature of today’s supply chains renders them inherently risky and creates increasing risk levels for multinational businesses, thus making supply chain transparency critical for performance yet complex to establish. Lack of effective visibility into their supply chains could potentially create blind spots for managers in their enterprise risk management structure; these blind spots in turn could cause substantial legal, financial, and reputational exposure. As a potentially promising solution, it is easy to get carried away by the promise a new technology like blockchain holds for its ability to streamline complex supply chain networks in order to improve the flow of information and make the networks more responsive [11]. Hence, knowing when to invest in it becomes critical. In the remainder of Section III, we discuss several practical and theoretical frameworks that can be utilized to determine whether a firm’s supply chain needs a blockchain.

A. Decision-Making Frameworks for Blockchain Adoption

While organizations should aim to achieve increased levels of information visibility and transparency, adopting a new technology like blockchain must be justified to help them achieve a strategic advantage in the market. The studies discussed below suggest decision-making frameworks that can be utilized for this purpose.

Several studies provide structured methodologies in the form of decision trees or a Business Model Canvas [12] – [15] that allows the identification and characterization of scenarios in which using blockchains is an adequate option. Another study [16] also assesses the suitability of blockchain in different areas

such as supply chains, electronic health records, identity data management, and stock markets by taking into account the technical characteristics of blockchain architecture to determine whether blockchain adoption is more beneficial as opposed to continuing using a conventional database. In addition, there also exist other studies [17], [18] detailing blockchain feasibility evaluation models, which also consider the costs versus benefits of increased energy utilization, consensus delay stemming from decentralization, and single point of entry for data storage [19] – [22].

Despite the many interesting potential uses of blockchain technology, not every process where blockchains can be implemented might need one if features such as peer-to-peer exchange, decentralization, or robust public record keeping have nothing more than an incremental financial advantage. Further, owing to scalability constraints an operation could present, it might not make sense to have every microtransaction recorded on a public blockchain. Therefore, blockchain implementation could be driven by the ability to establish an equilibrium between “economic efficiency through centralization and trustless exchange through decentralization [23]”.

As it relates to supply chains, there are currently several companies that provide blockchain solutions, such as Provenance [24], Everledger [25], IBM-Maersk [26], Skuchain [27], and Walmart [28], to name a few. Supply chain practitioners foresee blockchain adoption paving the way for *demand chains*, also referred to as *pull* [29], as opposed to a *supply chain*, also known as *push* [29].

The World Economic Forum has also published blockchain assessment questionnaires (see Fig. 1) to help organizations decide if a blockchain technology suits their business requirements.

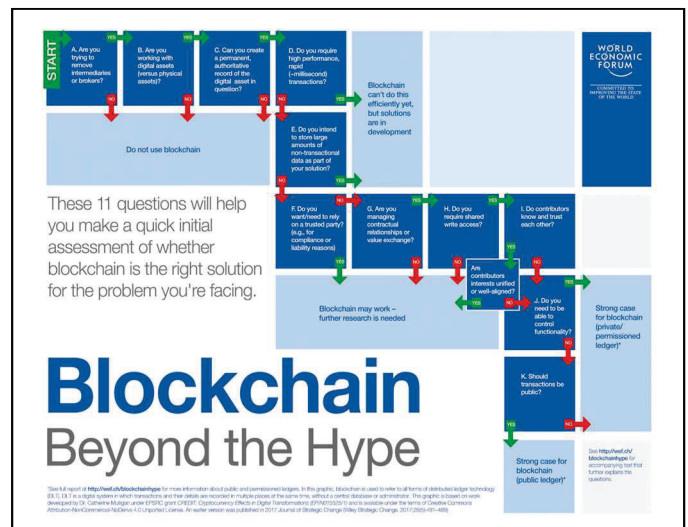


Fig. 1. A questionnaire to determine the suitability of blockchain as an appropriate technical solution for a problem (Creative Commons/World Economic Forum).

Although the cryptographically secure distributed ledger offers immutability and transaction safety, assessing the suitability of blockchain as a solution for information flow across supply chains is still a prerequisite, which is adequately fulfilled by the questionnaire.

B. Theoretical Frameworks for Blockchain Adoption

Blockchain technology has the potential to be a game changer in the supply chain domain. Although blockchain is often referred to a “disruptive” technology, labeling it a foundational technology is more appropriate, owing to its ability to create new foundations for socio-economic systems [30]. Hence, the importance of academic research that investigates its potential implications by establishing theoretical groundwork has an important role in rationalizing the proliferation of blockchains and in studying their as-of-now unexplored benefits to supply chains through rigorous research methodologies.

While the best practices in supply chain risk management have been extensively studied by academics as well as industry practitioners, these best practices can be summarized in the form of a few steps [31], [32]: identifying risk, developing a plan of mitigation, and exposure (ex-ante) and impact (ex-post) of the risk. The effectiveness of these steps is contingent upon availability of accurate information about the risk. A couple of terms that supply chain professionals encounter often in this regard are supply chain visibility and supply chain transparency. Visibility assists market efficiency by providing insight into data, thus making it available to stakeholders. Transparency allows firms to see through all the nodes and evaluate the entire supply chain, thereby assisting strategy fulfillment and social responsibility. This in turn drives communication and accountability. Transparency goes beyond visibility and requires deeper insight into a supply chain’s processes and a complex analysis of all its activities. In practice, although visibility and transparency are simply descriptors for flow of specific kinds of information, both are essential for supply chain operations. Blockchains can lead to improved supply chain visibility and optimization, thus leading to more effective risk management measures [33]. Since implementing blockchain technology is a means of improved information availability, we can reason that its managerial implications are better envisioned through the lens of foundational theoretical groundwork [34] – [38]. Theories pertinent to information sharing that could serve our interests well are the following: (1) Agency theory [39] – [41]; (2) network theory [42] – [44]; (3) resource-based view [45] – [47]; (4) transaction cost theory [48] – [50]; and (5) game theory and asymmetric information [51] – [54].

In order to understand better the potential implications of adopting blockchains in supply chains, and to empirically test, verify, and thus quantify the benefits of blockchain implementation, it is imperative to develop theoretical frameworks by adapting pre-existing theories. Such a framework could allow us to investigate the following questions: (a) How do blockchains affect the principal-agent relationship? (b) How do blockchains affect the nature and quality of business relationships within a supply-chain network? (c) How can blockchains be leveraged for gaining sustained competitive advantage? (d) How do blockchains affect the transaction costs on the organizational scale? (e) What role can a game-theoretic approach (cooperative and noncooperative games, games with asymmetric information, signaling) play in investigating the benefits of blockchain-enabled supply chain transparency?

Large and complex supply chains are representative of the principal-agent problem. *Smart Contracts* have gained traction in supply chain management owing to their facilitation being an underlying attribute of blockchain technology, as well as their potential for enabling trust in supply chains. Supply chain contracts are complex and dynamic agreements, with “regulatory and logistical constraints owing to them often transcending jurisdictions [55].” The importance of information exchange in a supply chain cannot be understated, and is at least as important as the flow of physical goods and services. Blockchains are expected to help integrate all participants within a supply chain network, and the audit trail in a blockchain ledger is expected to improve transparency across the supply chain. Smart contracts can improve supply chain visibility and efficiency, while also enabling provenance tracking by a wider range of stakeholders [55].

The prerequisite for achieving supply chain visibility is the need for all parties within the supply chain to exchange data. The effectiveness of this depends upon technical cooperation and technological sophistication. In order to critically assess the impact and limitations of blockchain in helping achieve the same, we require frameworks and models grounded in concepts borrowed from the aforementioned traditional theories. Academically sound and well-reasoned findings based on rigorous research methodologies will benefit decision making for all blockchain adopters.

IV. CURRENT RESEARCH GAPS

Investments worth upwards of \$1 billion [56] being made in blockchain startups as well as by large companies like IBM and Google are a clear indication of blockchain’s perceived potential. The fintech sector spends around \$1.7 billion annually on experimentation to determine the suitability of blockchain adoption [56]. Despite the increasing prominence of blockchain technology, there exist few peer-reviewed academic papers pertaining to operations and supply chain management; most existing research is featured in industry white papers, studies conducted by prominent management consulting firms, conferences, symposiums, and workshops. Larger number of peer-reviewed scholarly publications will legitimize blockchain adoption. Existing papers have studied the possibility of leveraging blockchains using smart contracts and IoT [32], [57], [58].

Several important research topics that needs to be addressed in the domain of operations and supply chain management can be categorized as follows:

1) *Designing efficient and responsive supply chains:* Leveraging blockchains to improve supply chain visibility for streamlining increasingly complex supply chains. The processes within a supply chain that are indicators of tangible benefits of improved visibility include the following:

a) *Reliable procurement* via transparent information sharing between various tiers of a supply chain. Inventory management is also a function of information sharing.

b) *Customer satisfaction* via improved service levels — this can be achieved through improved planning, quality, and compliance standards.

c) *Cost optimization* via operational visibility and transparent costs, such as freight costs, across the entire supply chain.

d) *Ethical supply chains* by leveraging the immutability and transparency attributes of a blockchain can encourage ethical business practices by empowering customers with an understanding of provenance, and reducing unfair and exploitative labor practices.

2) *Supply chain risk management*: End-to-end transparency within a supply chain allows monitoring risks by analyzing data and information pertaining to disruptions on either the supply or the demand side.

3) *Privacy, trust and security, consensus building*: Supply chain privacy and security issues exist, in the form of sourcing or transactions details being public and the distributed ledgers being susceptible to cyber attacks. If the firms that form the supply chain do not own the blockchain, then data ownership is transferred to a third party, the party being the blockchain provider. Proof-of-concepts to mitigate these challenges is a topic that is commonly under consideration amongst academics and organizations wanting to adopt blockchain technology.

V. SOLUTION APPROACH/METHODOLOGY

We can argue that blockchain is a novel tool for implementing solutions based on traditional theories and methodologies in an efficient way, and at a broader scale. Blockchain implementation in supply chains requires engagement across the entire chain such that every actor within the chain must be willing to adopt and use the technology. Blockchain adoption is also contingent upon widespread connectivity and digital literacy. In this section, we discuss a few possible approaches towards realizing the research questions we pose in the prior section.

A. Blockchain and Game Theory

Blockchains in supply chain have the potential to establish a Nash Equilibrium where no one participant within the supply chain setup has an incentive to monopolize data.

A segment of the healthcare industry supply chain, specifically insurance and healthcare provision, is a characteristic example of prisoner's dilemma. From a game-theoretic standpoint, this application can be viewed as a two-person game between insurance providers and payers, wherein both players behave sub-optimally. While on the one hand, the provider is better off providing low quality service, regardless of whether the payer chooses to pay for a claim; alternatively, the payer is better off rejecting the claim, regardless of the quality of the service provided [59]. As a result, a negative social cost is incurred in the absence of incentives for both players. On the blockchain however, since all actions are permanently recorded, providing payment or good service is deemed economically beneficial, thus incentivizing players to follow socially optimal strategies.

Blockchains have a legitimate case for useful applications in the domains of both financial and governance supply chains. Lack of trust among financial players in the banking sector results in a suboptimal equilibrium owing to the lack of

transparency in transactions. The resulting systemic inefficiencies have in the past led to financial contagion. Integrating blockchain in traditional banking can help improve trust among market participants by improving transparency in order to increase the reliability of banking information. Blockchain adoption for effective governance can be considered to be an example of a coordination game [60].

Blockchain adoption for improving transparency in order to achieve social optimum during energy sharing and trading between distrustful players within a community attempting to adopt renewable energy sources such as solar panels or other decentralized energy storage units could be modeled as a non-cooperative game [61], Stackelberg game [62], or an auction model [63], [64].

In summary, a game-theoretic approach can be used to examine other real-world supply chain settings that are subject to heightened social awareness, and investigate whether blockchains can advance the incumbent setup by incentivizing socially beneficial actions.

B. Blockchain and Grounded Theory

The diversity of global supply chains renders consumers unaware of the provenance of the products they buy. Rising social awareness on the part of consumers about the impact of environmentally unsustainable and unethical (ex. human rights and labor conditions) practices means that the desire for transparency within supply chains has never been larger. The supply chain disruptions due to reputational risks incurred from such practices lead to implications that are difficult to recover from. Stock prices are negatively impacted, to the tune of an average of nine percent [65], [66]. Governments too are being increasingly challenged to understand and manage risks across supply chains by introducing legislations to ensure ethical practices in supply chains. Blockchains can increase efficiency, reduce costs, and promote transparency, thereby driving social impact by making supply chains both responsible and resilient. Therefore, when exploring the role and feasibility of blockchains in an academic setting, we should compare the effectiveness of blockchain implementation to the more traditional information management and signaling mechanisms. It is critical that the transformative ability of blockchains to allow companies to make a positive social impact must drive the development of blockchain applications.

One way to motivate studies that investigate the feasibility of leveraging blockchains for improved transparency and visibility is to justify the use of blockchains in an industry. Gauging market reception to announcements of blockchain can be studied by conducting an event study around these announcements. Looking at the data of patents that have been filed by US based publicly listed companies (see Fig. 2) [67], it is no surprise to find that most firms are either financial firms, large technology or service operations companies with globally diverse and complex supply chains, or major technology solution providers to companies in the retail and logistics sector.

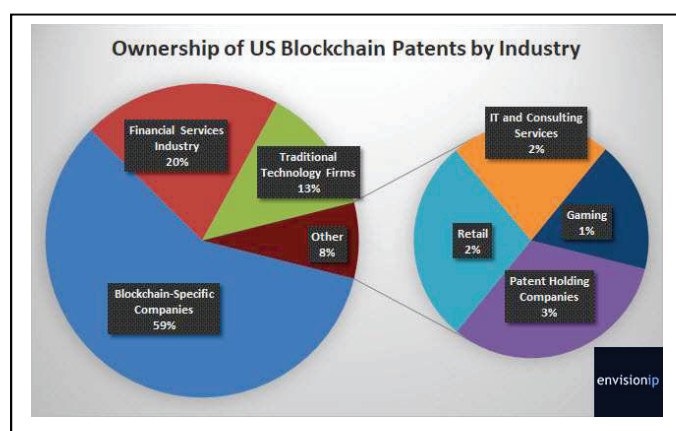


Fig 2. Industry sectors comprising list of publicly-traded companies with the most blockchain patents (until December 2018) (Source: www.patentvue.com)

Examples of industry sectors whose supply chain practices are subject to a higher level of scrutiny on a societal level include the following [68]: agriculture; fashion; conflict minerals; humanitarian aid and philanthropy; pharmaceuticals; cosmetics and personal care; energy generation and distribution; and, healthcare. We can posit that these supply chains would benefit from increased transparency and visibility via blockchain adoption, which in turn would act as an affirmative signal for consumers of the products and services of the aforementioned industries. We can motivate researching blockchain implementation within these industries to address some of the research questions we have put forth in the prior section, namely, *designing efficient and responsive supply chains* and *supply chain risk management*. This can be considered to be an attempt towards motivating a framework for the academic investigation of the feasibility and usefulness of blockchain adoption for social good within supply chains.

In addition, industry practitioners have collaborated with prominent consulting firms for long for studying blockchain adoption. Therefore, it is important to combine academic ideas with the prevalent practice. For instance, combining some of the ideas presented previously in this section with other pre-existing frameworks [69], [70] can allow us to assess the security risks, the need for and levels of trust within multiple parties, and concerns about levels of market adoptions and the prevalent market regulations associated with implementing blockchain.

VI. CONCLUSION

Blockchain technology has the potential for transforming the traditional supply chains by increasing the efficiency of information flow within them, thereby making them more responsive and resilient. On the other hand, a few obstacles [6], [71] to a more widespread blockchain adoption include lack of suitable use cases, governance of the technology, lack of regulation, security concerns, and scalability and performance concerns. Therefore, the one consensus academics and practitioners can arrive at without a doubt is that more descriptive and prescriptive research is needed before this technology acquires legitimacy and becomes ubiquitous as TCP/IP and Internet once did.

Studies motivated by theory that investigate whether better information sharing within a supply chain, enabled by

blockchain technology, can lead to better risk management and disruption mitigation capability would contribute to the much needed yet currently lacking academic literature. This study attempts to motivate pertinent research questions and a few possible solution approaches in order to study blockchain implementation in supply chains in a manner befitting traditional academic rigor and nuance.

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